Robotics as a Career

By Peter I. Corke

In this issue, I am talking with John J. Craig (JJC), vice president of EponaTech LLC, about life, robots, and books.

PIC: **How do you get to be in robotics? Can you recall the day, the moment that you decided robotics was for you?**
JJC: I can. My older brother built a robot for his high-school science project (he was in 12th grade, and I was in eighth). It was a crazy 6-ft tall wooden box on wheels kind of robot with a tape recorder inside it so that it could talk. I thought it was so cool that I decided that when I got to high school I would build a better one. Sure enough, my 11th grade science project was a robot that had a four degree of freedom (DoF) arm, photocells for eyes, and of course, a tape recorder inside it so that it could talk. I then went off to college temporarily forgetting about robots, as I was sure that nobody could earn a living in the real world building robots. But five years later, after a master's degree, I discovered that robots were becoming a reality in manufacturing—so I said "that's it!"—I was set to do robotics for a career.

PIC: **So where did you go to college and what did you study?**
JJC: I went to RPI in Troy, New York, for undergraduate study and stayed for a fifth year to get a master's degree. Both degrees were in electrical engineering. In those days, the one and only robotics project on campus

John Craig and one of his horses, Willow.

was a Mars Rover project funded by NASA's Jet Propulsion Laboratory (JPL). In my fifth year at RPI, I was one of the coleaders of that project. This rover was featured in an episode of the television series *Cosmos* by Carl Sagan. After graduating from RPI, I took a job at JPL. I began my involvement with robotics research there, stayed two years, and then went off to Stanford to pursue a Ph.D. degree.

PIC: **What robots did you work on at JPL?**
JJC: The JPL had a 6 DoF electric arm made by Vic Scheinman. It was the third arm that he made—the first two were the blue and yellow arms at the Stanford Artificial Intelligence Laboratory. I had a good fortune to be hired by Marc Raibert at JPL and work with him for my first year there. Marc had gotten one of the first force-sensing wrists from Vic Scheinman and wanted to do something with it. We developed the hybrid position/force control strategy and did the first experiment using it with the Scheinman arm: we put a peg into a hole using active force control, which was a big news at that time. During my second year at JPL, I worked with Carl Ruoff, who was also a great teacher. Carl had done a lot of robotics work at Bendix Corporation, but as it was proprietary, it wasn't published, and nobody knew about it—but it was great stuff—two arms cooperating and with vision—way back in the mid-1970s. In the second year at JPL, working with Carl, I developed a robot control language and system that we called *JPL autonomous robot system* (JARS).

PIC: **So you're one of those lucky people to have a method named after them: the Raibert-Craig method. What was the motivation for going to Stanford, JPL at that time sounds like a pretty exciting place to work.**
JJC: At JPL I really got the feel for, and love of, doing research. A lot of people at JPL had Ph.D. degrees, and while I had never previously thought about pursuing a Ph.D. degree, with some encouragement from my wife, I started to consider it. Then I found out that Stanford had more robots and better computers than JPL. So, it felt like an upgrade for my research environment. Still it was a difficult choice, and also I nearly went instead to Massachusetts Institute of Technology (MIT) or Purdue. Lou Paul was at Purdue at the time, and I was interested in working with him.

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newest links, see http://www.youtube.com/watch?v=3CR5y8qZf0Y&feature=player_embedded from RAS member Raffaello D'Andrea's laboratory at ETH Zurich, which shows two autonomous flying robots enjoying a bit of tennis.

New RAS Chapters

We are happy to welcome several RAS Section Chapters and student branch Chapters, which have been recently approved (see tables on previous page). To organize a new RAS Chapter in your Section or school, see http://www.ieee.org/about/volunteers/tab/creating_a_chapter.html#sect2 (new Section Chapter) or http://www.ieee.org/documents/2009_branchchapterpetition.pdf (student branch Chapter).

Turning Point

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PIC: Tell me about Stanford, who you were working with, and how you got from compliance to adaptive control?

JJC: Stanford was a great place, and I was fortunate to work with, and learn from, Ken Salisbury, Oussama Khatib, Bernie Roth, Jeff Kerr, Tom Binford, and many others. Meeting Oussama Khatib, I realized that his work was the final missing piece to the hybrid force controller, since his way of handling dynamics could bring the full dynamic decoupling needed to make the hybrid force-controller highly performant. So he and I worked on that. With Ken Salisbury, I wrote the control code that was the first to wiggle the fingers of his three-fingered hand, and he and I wrote a paper that won the Best Paper of the Conference Award at the American Control Conference. I started as a student of Tom Binford's, but all my involvement with mechanical types got me more and more to the mechanics and control aspects, and partway through my Stanford years, I switched to Bernie Roth as my thesis advisor. The thesis work ended up being on adaptive control, and what I wanted to do was to add an adaptive aspect to a controller that fully decoupled the mechanism by means of a dynamic model, either in joint or Cartesian spaces. The adaptive element would just tune-up the parameters appearing in the dynamic model used by the control computer. Every other adaptive controller present before I started this work used adaptation to avoid computing a complete dynamic model, but computers were now fast enough to compute the model, so I thought that the adaptation should be used to learn the correct parameter values appearing in that model. I still think it's the way to build a highly performant adaptive controller.

Silma created an offline programming tool for industrial robots.

PIC: So how do you top that experience?

JJC: Hard to top it, it was a lot of fun, and I learned a lot.

PIC: When did you write the textbook?

JJC: Bernie Roth was teaching the robotics course at Stanford, and when he was going away for a year on sabbatical, he asked if I wanted to teach it while he was gone. I was sufficiently nervous (teaching my first course) that I decided to rather laboriously develop detailed notes to hand out in class—and by writing out everything, it helped me get my teaching act together. I had time to do it, because Bernie asked me to teach about nine months before the class was to start. At the end of the course, the notes formed the core of the book. But it took two or three more years to turn the notes into a book. I did that while being a graduate student and working on the thesis too. So both the book and thesis came out in 1986.

PIC: So the book on adaptive control came out after this one? That was a pretty productive period of writing. What came next?

JJC: The first textbook was pretty exciting to Addison-Wesley (the original publisher) at the time, and they actually pursued me to write another or a follow-on book. I suggested that my Ph.D. thesis could be turned into a book, and they wanted to do that. We knew that it would be a very different kind of book, more like a monograph, and would not go to as large a market as my *Introduction to Robotics* textbook. In the end, the second book was literally almost a copy of my thesis—only a few words got changed. As predicted, it was for a rather specialized audience, and went out of print a few years later whereas the first textbook has gone on to a second and now a third edition and continues to be used today. Both the textbook and thesis (obviously) were finished before I graduated from Stanford.

While still finishing these up, I was also involved in a start-up company in a Silicon Valley called *Silma*. It had been my plan to go with them full-time after graduating, and I ultimately did, but only after a brief flirt with the academia. The University of California at Berkeley offered me a tenure track position, which I really hadn't pursued, so I was surprised by the offer and sufficiently flattered that I accepted. But soon afterward realized that I did want to do the Silicon Valley start-up after all, and so, regretfully, left Berkeley really before I ever got started there.

PIC: What did Silma do, and what did you do there?

JJC: Silma created an offline programming tool for industrial robots. One could, for example, bring in the three-dimensional (3-D) computer-aided design (CAD) model of a car body and investigate where to place the robot so that it could reach all the spot-weld points, how the spot-weld gun should be mounted on the flange, work out all the details, make sure the selected robot could do the job, avoid collisions, and so on. Then, the offline programming system would create an online program and download it to the actual machine. To make all this actually work, it got me into the problem of robot calibration, and of course, also learned a lot about 3-D modeling, graphics, and fast collision detection algorithms. Our timing was good, and when we started, graphics were pretty wimpy, and each link of the robot was depicted as a single line segment. But within a year or two, the first Silicon Graphics workstations were out, and we had beautiful 3-D surfaced models of the robot links, the tooling, and the work pieces. We were at the right time to put together practical systems that tied CAD models to automated manufacturing. I was convinced at that time that we really didn't need faster or more-accurate robots; we needed easier-to-use robots so that they could be cost effectively deployed into factories. So, that's what we tried to do.

PIC: How did you come to work for Adept?

JJC: Adept acquired Silma in 1995, and we became the Silma Division of Adept. I got involved in some robot design issues for the main business of Adept, but mostly I worked in the Silma division and created a new version of the offline-programming and simulation system that was specific to the sort of applications that Adept robots were used in. I was happy to work with Brian Carlisle, Bruce Shimano, Rick Casler, and others. Silma existed for about ten years before being acquired by Adept, and I stayed with Adept for an additional seven years after the acquisition, so all together, I spent 17 years in the area of CAD-driven offline programming of robots.

PIC: What came next?

JJC: After Adept, I worked for a high-tech company called *Invenios*. They make high-precision robots—generally intended for work on very small-sized applications—to assemble fiber optics, as stages for atomic-force microscopes, and other small things. One project there was a 6-DoF parallel robot for which I had to develop an iterative kinematic algorithm, because the geometry was not closed-form solvable. It could position within tens of micrometer and had a workspace smaller than a cubic centimeter. I also developed a digital signal processing (DSP)-based encoder that was pretty advanced.

PIC: Can you tell me what you are up to now?

JJC: For the past two years, I have been an entrepreneur—my wife and I ran our two companies: one is a software company (www.Metron-Imaging.com) that makes a software package used for veterinary radiography; the other company is a horseshoe company (www.EponaShoe.com), and we make polyurethane horseshoes that are better for the horse than the 2,000-year-old metal technology still widely used. Our shoes were on the top dressage horse on the U.S. Olympic team in Beijing. It may sound like a big change for me, but basically I've just shifted from robot mechanics to the related field of biomechanics, with a special focus on the horse. We've owned horses for 20 years and have 12 horses now, and so, over time I guess I got interested in how they work.

PIC: Finally, what do you see as the big opportunities and challenges for robotics today?

JJC: I'm pretty excited about all the work in the area of haptics and robots than can control their applied forces and work safely alongside humans. I think there are more applications to come in that area.

> **Markets for robots would further open up if they were less costly to deploy.**

I guess I would also still say (as I did 25 years ago) that making robots easier to use is a key thing. This means either they become way easier to program or they become more autonomous. Markets for robots would further open up if they were less costly to deploy—and that comes from them being easier to use.

ERRATA

In "RAS Finances," *IEEE Robotics & Automation Magazine*, volume 18, number 1, March 2011, pp. 8–10, the text at the end of the first paragraph of the "Budget Year 2011 and Summary" section should have been "In the past several years, the budget surpluses were −$78.9K, −$46.2K, −$42.5K, −$59.1K, $78.1K in 2006, 2007, 2008, 2009, and 2010, respectively."

These surplus numbers appeared incorrectly; they inadvertently had been shifted by one decimal place. For example, −$46.2k was printed as −US$462,000. We apologize for the error and any confusion it may have caused.

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